第七讲 扭转变形
非圆杆扭转、
薄壁杆扭转

作业：4-16,22,26,27,36,9*,23*

第四章　扭转

4.1 扭力偶矩与扭矩

4.2 圆轴扭转应力

4.3 圆轴扭转强度条件与合理强度设计

4.4 圆轴扭转变形与刚度条件

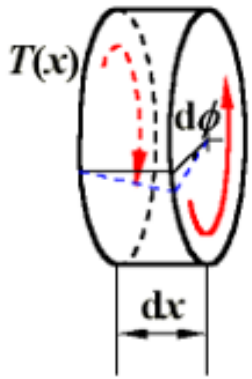
4.5 圆轴扭转静不定

4.6 非圆截面杆件扭转

4.7 薄壁杆扭转

上讲回顾

扭转平面假设



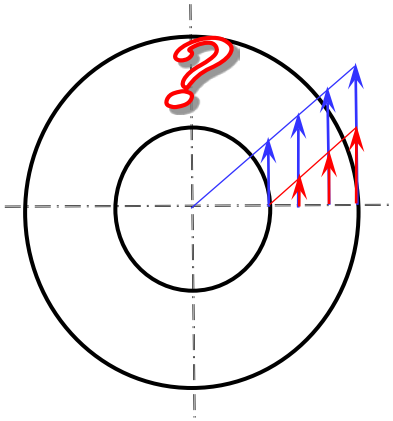
$$\gamma_\rho \mathrm{d}x = \rho \mathrm{d}\varphi, \quad G\gamma_\rho = \tau_\rho, \quad \int_A \rho\tau_\rho \mathrm{d}A = T$$

$$\boxed{\mathrm{d}\varphi = \frac{T}{GI_P}\mathrm{d}x}$$

$$\tau_\rho = \frac{T\rho}{I_\mathrm{p}} \qquad \tau_{\max} = \frac{T}{W_\mathrm{p}} \le [\tau]$$

$$I_\mathrm{p} = \frac{\pi D^4}{32}\left(1-\alpha^4\right)$$

$$W_\mathrm{p} = \frac{\pi D^3}{16}\left(1-\alpha^4\right)$$



平面假设揭示了
微段变形的基本特征

第七讲 扭转变形
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第四章 扭转

4.4 圆轴扭转变形与刚度条件

1、圆轴扭转变形

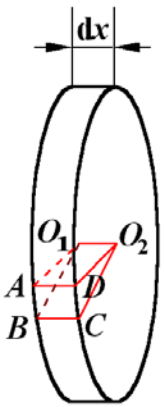
2、圆轴扭转刚度条件

3、例题

1、圆轴扭转变形

圆轴微段扭转角

$$\mathrm{d}\varphi = \frac{T(x)}{GI_p(x)}\mathrm{d}x$$

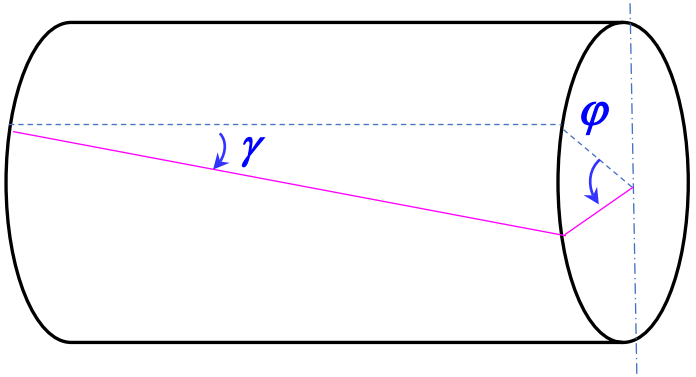


圆轴扭转角

$$\varphi = \int_l \frac{T(x)}{GI_p(x)}\mathrm{d}x$$

GI_p—圆轴截面扭转刚度，简称扭转刚度

对于常扭矩、等截面圆轴



$$\varphi = \frac{Tl}{GI_p}$$

$$\Delta l = \frac{F_N l}{EA}$$

$$R\varphi = \frac{TR}{GI_p}\, l = \gamma_{\max} l$$

扭转角与切应变相关

2、圆轴扭转刚度条件

$$\theta = \frac{\mathrm{d}\varphi}{\mathrm{d}x} = \frac{T}{GI_{\mathrm{p}}}$$

$$\left(\frac{T}{GI_{\mathrm{p}}}\right)_{\max} \le [\theta]$$ 圆轴扭转刚度条件

$[\theta]$ - 单位长度的许用扭转角

- 一般传动轴， $[\theta] = 0.5° \sim 1°/\mathrm{m}$
- 注意单位换算：

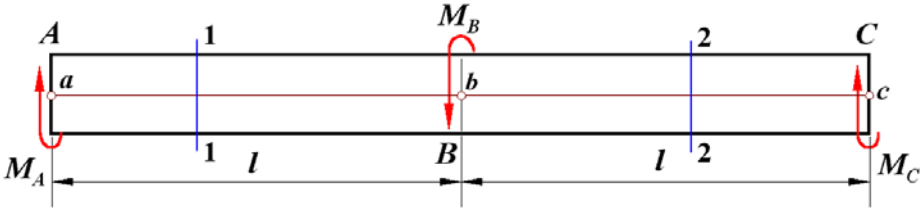
$$1\ \mathrm{rad/m} = \frac{180°}{\pi}\ /\mathrm{m}$$

3、例题

例4-5 已知 M_A=180N·m，M_B=320N·m，M_C=140 N·m，I_p= 3×10^5 mm^4，l = 2 m，G = 80 GPa，[θ] = 0.5°/m 。1） φ_{AC}=? 2）校核轴的刚度

解： 1）变形分析

扭矩设正



$$T_1 = M_A = 180\ \mathrm{N\cdot m}$$

$$T_2 = -M_C = -140\ \mathrm{N\cdot m}$$

$$\varphi_{AC} = \varphi_{AB} + \varphi_{BC} = \frac{T_1 l}{GI_p} + \frac{T_2 l}{GI_p} = 1.5\times10^{-2} - 1.17\times10^{-2} = 0.33\times10^{-2}\ \mathrm{rad}$$

2）刚度校核

$$\left(\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right)_1 = \frac{T_1}{GI_p}$$

$$\left(\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right)_2 = \frac{T_2}{GI_p}$$

$$\because\ |T_1| > |T_2|$$

$$\therefore\ \left(\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right)_{\max} = \frac{\varphi_{AB}}{l} = \frac{1.5\times10^{-2}}{2}\frac{180}{\pi} = 0.43\ ^\circ/\mathrm{m} < [\theta]$$

轴的刚度合格

第四章 扭转

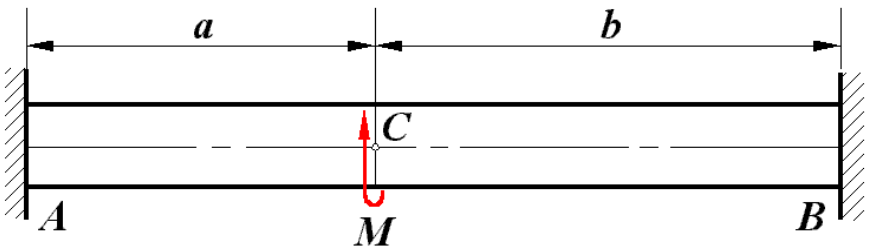
4.5 圆轴扭转静不定

1、圆轴扭转静不定问题分析

2、例题

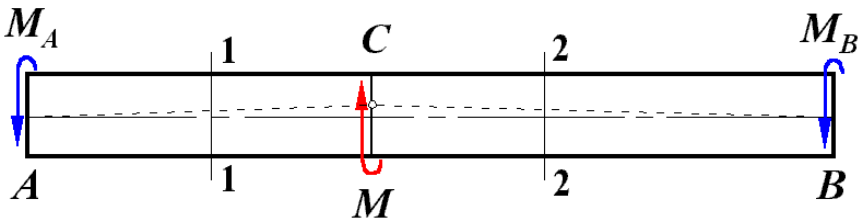
1、圆轴扭转静不定问题分析

两端固定圆轴，中间受外力矩作用，求支反力偶矩



问题分析

综合考虑静力学、几何、物理三方面



$$M_A + M_B - M = 0$$

$$\varphi_{AB} = \varphi_{AC} + \varphi_{CB} = 0$$

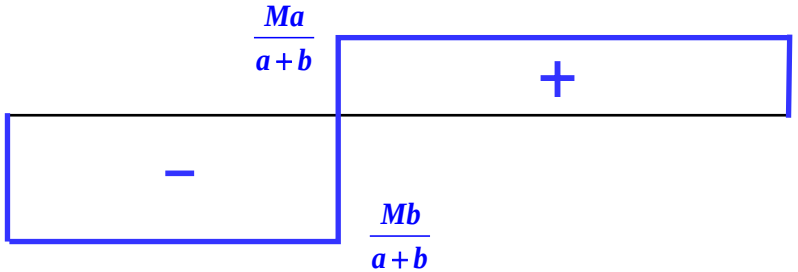
$$\varphi_{AC} = \frac{T_1 a}{GI_p} = \frac{(-M_A)a}{GI_p}$$

$$\varphi_{CB} = \frac{T_2 b}{GI_p} = \frac{M_B b}{GI_p}$$

一度静不定

$$M_A = \frac{Mb}{a+b}$$

$$M_B = \frac{Ma}{a+b}$$

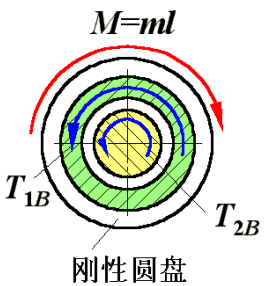


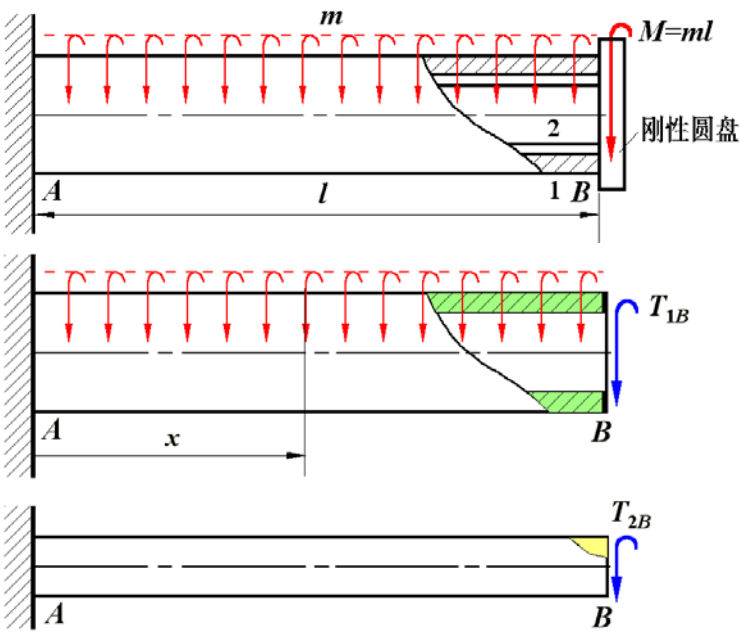
内力按刚度分配

由 $T = \frac{GI_P}{l}\Delta\varphi$ 可见：$\frac{GI_P}{l}$ ——产生单位扭转角所需要的力矩，圆轴的刚度（系数）

2、例题

例4-6 图示组合圆轴，承受集度为 m 的均布扭力偶，与矩为 $M = ml$ 的集中扭力偶。已知：$G_1 = G_2 = G$，$I_{p1} = 2I_{p2}$。试求：圆盘的转角。





解： 1）解静不定

$$T_{1B} + T_{2B} = ml \qquad \boxed{\varphi_{1B} = \varphi_{2B}}$$

$$\varphi_{1B} = \int_0^l \frac{T_1(x)}{G_1 I_{p1}} dx, \qquad T_1(x) = T_{1B} + m(l - x)$$

$$\varphi_{2B} = \frac{T_{2B} l}{G_2 I_{p2}}$$

$$T_{1B} = T_{2B} = \frac{ml}{2}$$

2）求转角

$$\varphi_B = \varphi_{2B} = \frac{T_{2B} l}{G I_{p2}} = \frac{ml}{2G I_{p2}}$$

3）思考：

若圆轴预先转动一角度，再用刚盘连接，如何分析？

第四章 扭转

4.6 非圆截面杆件扭转

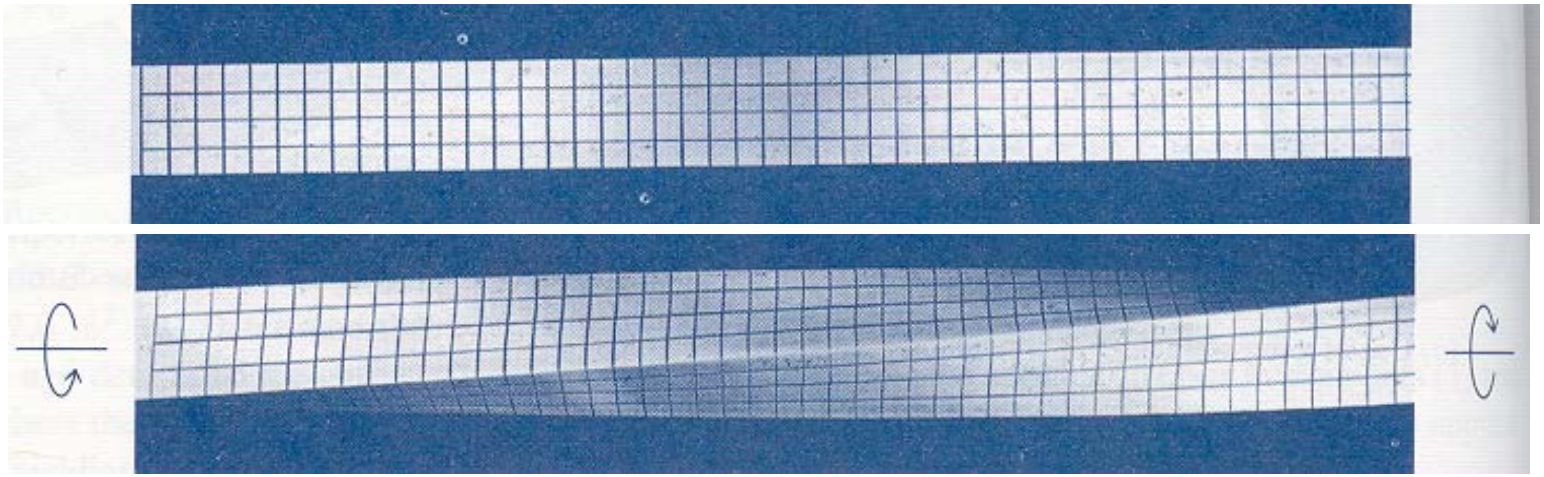
1、自由扭转和限制扭转

2、矩形截面杆件扭转

3、椭圆等非圆截面杆件扭转

1、自由扭转与限制扭转

非圆截面杆扭转



试验与分析表明

平面假设不成立

截面翘曲

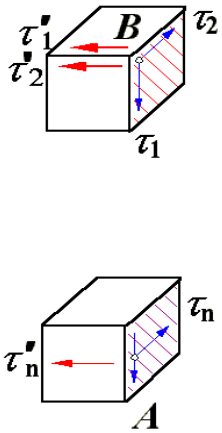
限制扭转

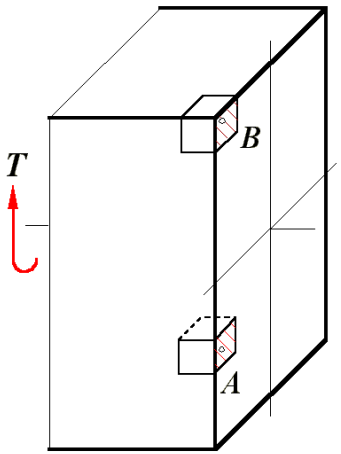
截面正应力

自由扭转

2、矩形截面杆件扭转

试验可见：截面翘曲，角点处 γ 为零，侧面中点处 γ 最大





截面应力分布特点

- 横截面边缘各点处，切应力 // 截面周边；
- 横截面上角点处，切应力为零
- 横截面周边长边中点处，切应力最大

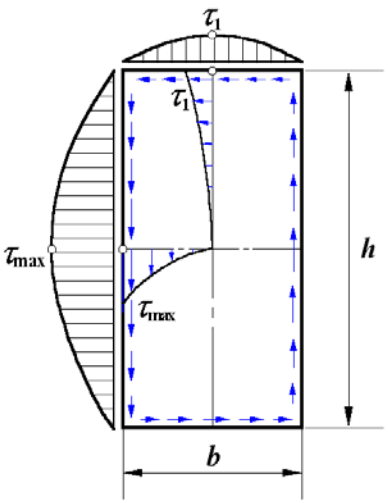
2、矩形截面杆件扭转

弹性力学解

1）长边中点处最大 $\tau_{\max} = \dfrac{T}{W_t} = \dfrac{T}{\alpha h b^2}$

2）短边中点处局部最大 $\tau_1 = \gamma \tau_{\max}$

3）扭转角 $\varphi = \dfrac{Tl}{GI_t} = \dfrac{Tl}{G\beta h b^3}$



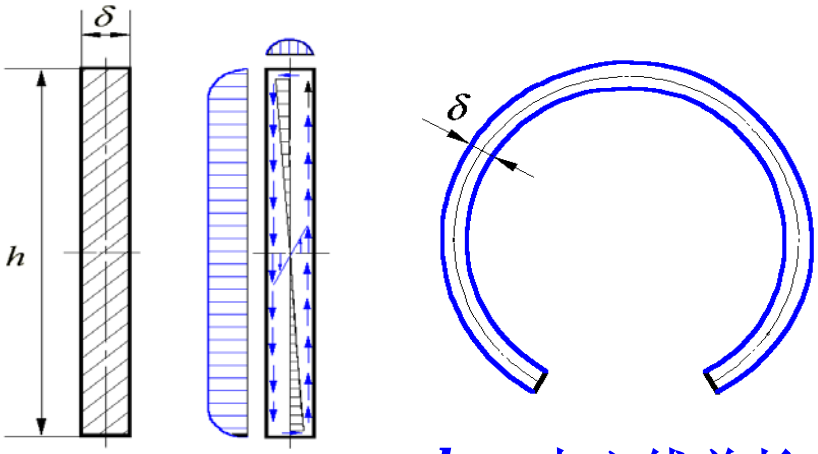
系数 α, β, γ 与 h/b 有关，见表4-1

狭长矩形截面杆件扭转

当 $h \geq 10\delta$ 时， $\alpha, \beta \to 1/3$

$$\tau_{\max} = \frac{3T}{h\delta^2} \qquad \varphi = \frac{3Tl}{Gh\delta^3}$$

最大应力与中心线形状无关

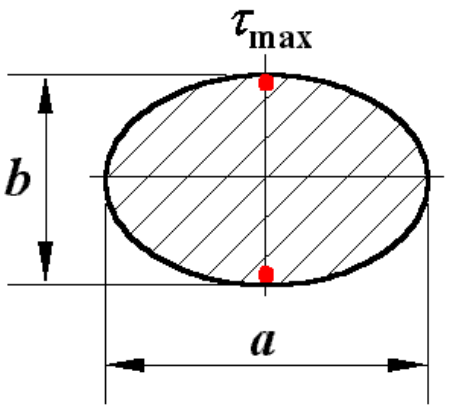


h - 中心线总长

3、椭圆等非圆截面杆件扭转

$$\tau_{max} = \frac{T}{W_t}$$

$$\varphi = \frac{Tl}{GI_t}$$



- W_t, I_t 的量纲分别与 W_p, I_p 相同
- 计算公式见附录D

第四章 扭转

4.7 薄壁杆件扭转

1、开口与闭口薄壁杆

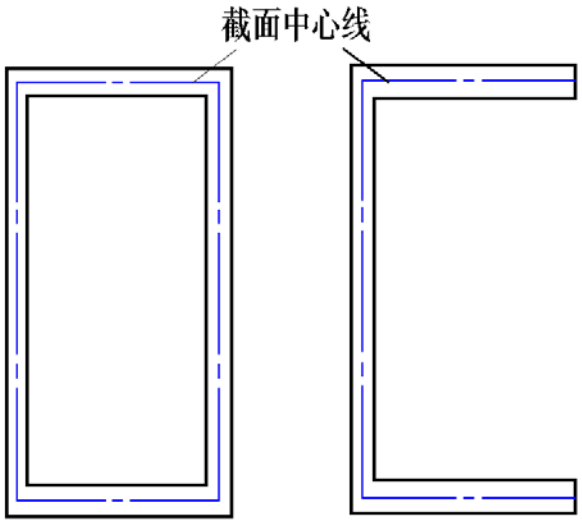
2、闭口薄壁杆扭转应力与变形

3、开口薄壁杆扭转分析

4、薄壁杆合理截面形状

1、开口与闭口薄壁杆

- 截面中心线—截面壁厚平分线
- 闭口薄壁杆—截面中心线为封闭曲线的薄壁杆
- 开口薄壁杆—截面中心线为非封闭曲线的薄壁杆

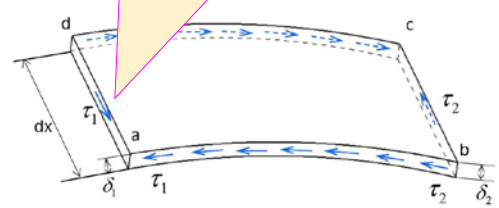


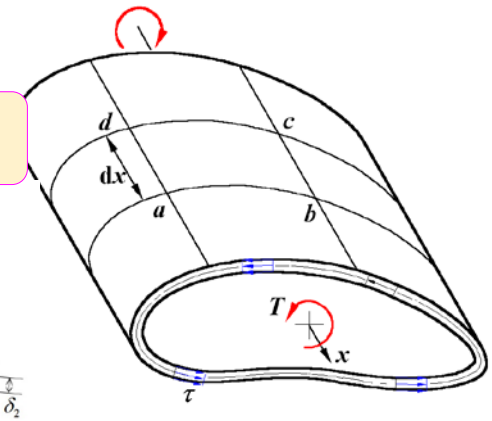
2、闭口薄壁杆件扭转应力与变形

假设：1）切应力沿壁厚均匀分布，

2）平行于中心线切线

与图4-28(b)不同



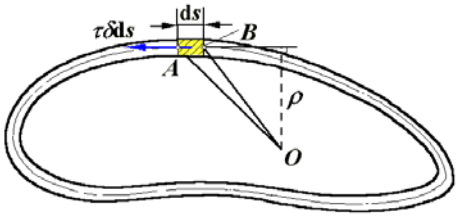


取单元体　　单元体平衡

$$\tau_1\delta_1 \mathrm{d}x - \tau_2\delta_2 \mathrm{d}x = 0$$

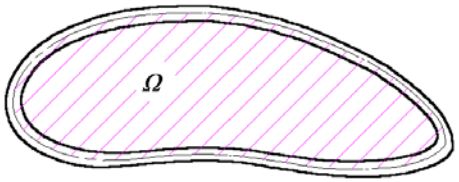
$$\Rightarrow \quad \tau_1\delta_1 = \tau_2\delta_2 \qquad\qquad \tau\delta = q$$

剪流：中心线单位长度上的剪力；在截面上为常数。



截面力矩平衡

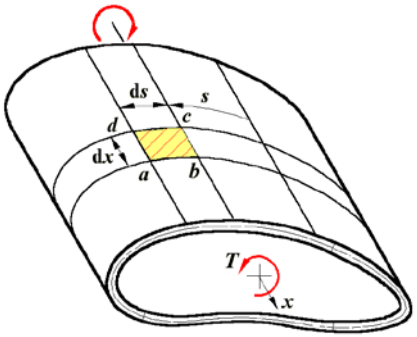
$$\oint \rho \cdot \tau\delta \mathrm{d}s = T \qquad \boxed{\tau = \frac{T}{2\Omega\delta}}$$

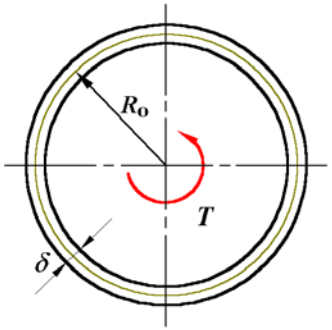


- 切应力与截面中心线所围面积Ω成反比；
- τ_{max}发生在壁厚最薄处。

2、闭口薄壁杆件扭转应力与变形

常扭矩、等截面、线弹性闭口薄壁杆的扭转角





单元体 *abcd* 的应变能

$$\mathrm{d}V_\varepsilon = \frac{\tau^2}{2G}\delta \mathrm{d}s\mathrm{d}x$$

闭口薄壁杆扭转应变能

$$V_\varepsilon = \int_l \oint \frac{T^2}{8\Omega^2 G\delta}\mathrm{d}s\mathrm{d}x = \frac{T^2 l}{2GI_t}$$

$$\frac{T\phi}{2} = \frac{T^2 l}{2GI_t} \quad \Rightarrow \quad \varphi = \frac{Tl}{GI_t}$$

$$I_t = \frac{4\Omega^2}{\oint \frac{\mathrm{d}s}{\delta}} \qquad I_p = \frac{4\left(\pi R_0^2\right)^2}{\frac{2\pi R_0}{\delta}} = 2\pi R_0^3\delta$$

薄壁圆管

$$\tau = \frac{T}{2\pi R_0^2\delta} \qquad \varphi = \frac{Tl}{2G\pi R_0^3\delta}$$

按空心圆计算

$$\beta = \frac{\delta}{2R_0}$$

$$\tau = \frac{T}{W_p} = \frac{T}{2\pi R_0^2\delta}\left(1 + \frac{\beta - \beta^2}{1+\beta^2}\right)$$

$$\varphi = \frac{Tl}{GI_p} = \frac{Tl}{2G\pi R_0^3\delta}\left(1 - \frac{\beta^2}{1+\beta^2}\right)$$

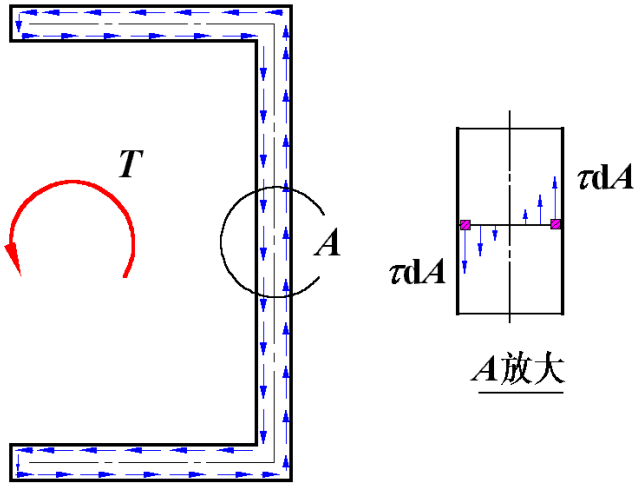
当 $2R_0/\delta>20$，薄圆管的解答足够精确

3、开口薄壁杆扭转分析

1）与狭长矩形截面相当，应力与中心线形状无关；

2）扭转切应力沿厚度非均匀，沿截面周边呈“环流”分布；

3）开口薄壁杆的抗扭性能极差。

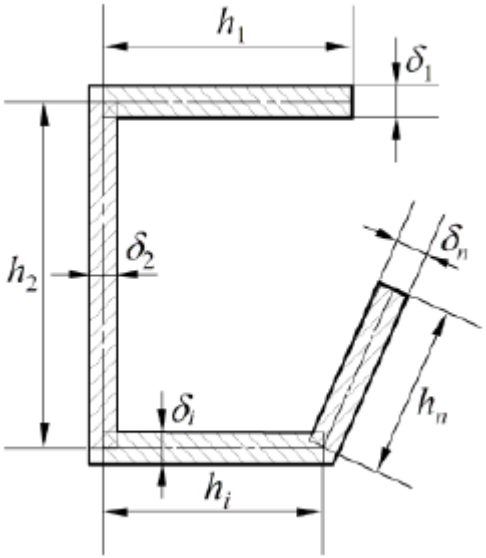


等厚度开口薄壁杆

$$\tau_{\max} = \frac{3T}{h\delta^2}$$

$$\varphi = \frac{3Tl}{Gh\delta^3}$$

h - 中心线总长

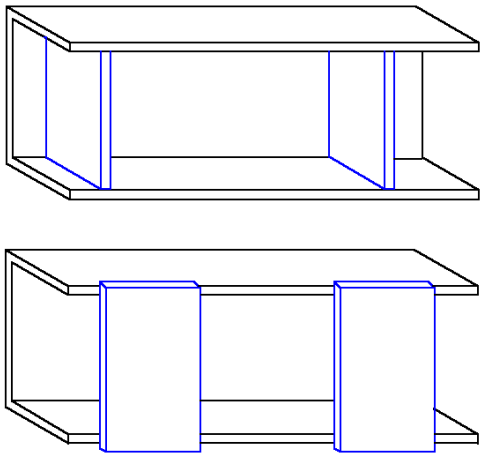


变厚度开口薄壁杆

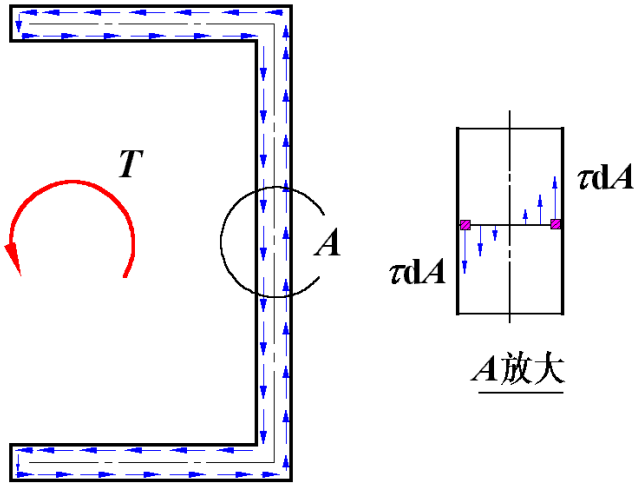
$$\tau_{\max} = \frac{3T\delta_{\max}}{\sum_{i=1}^{n} h_i\delta_i^3}$$

$$\varphi = \frac{3Tl}{G\sum_{i=1}^{n} h_i\delta_i^3}$$

3、开口薄壁杆扭转分析



采用隔板或肋板，将显著提高开口薄壁杆的抗扭性能



思考：

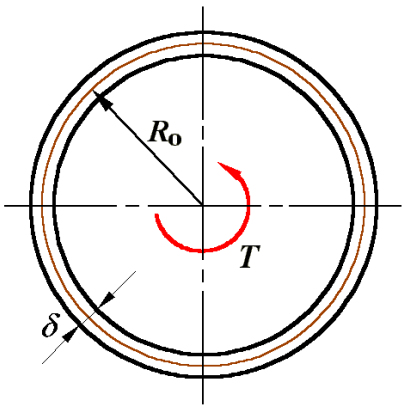
为什么不能假设切应力沿厚度均匀分布？

截面剪力为零

4、薄壁杆合理截面形状

$$\tau_{闭}=\frac{T}{2\pi R_0^2\delta}\qquad \tau_{开}=\frac{3T}{h\delta^2}=\frac{3T}{2\pi R_0\delta^2}$$

$$\varphi_{闭}=\frac{Tl}{2G\pi R_0^3\delta}\qquad \varphi_{开}=\frac{3Tl}{Gh\delta^3}=\frac{3Tl}{2\pi R_0 G\delta^3}$$



1）闭口远比开口好，开口杆不能承扭！

2）等壁厚好于变壁厚；

3）中心线长相同的情况下，

圆形好于非圆形，正方形好于矩形。

$$R_0=10\delta$$

$$\frac{\tau_{开}}{\tau_{闭}}=3\left(\frac{R_0}{\delta}\right)=30$$

$$\frac{\varphi_{开}}{\varphi_{闭}}=3\left(\frac{R_0}{\delta}\right)^2=300$$

本讲：圆轴扭转变形、

非圆截面杆件扭转

开口薄壁杆件不能承扭

下讲：剪力弯矩图